UNIT 4
INTERFACING

Basic Description of the 8255

- The 8255 provides **24 I/O lines** which may be individually programmed in **2 groups** of 12 I/O lines and used in **3** major **modes of operation**.
- These 24 I/O lines organized as **three 8-bit I/O ports** labeled **A**, **B**, and **C**.
- The chip interfaces directly to the data bus of the processor, allowing its function to be programmed;
- That is, in one application a port may appear as an output, but in another, by reprogramming it, as an input.

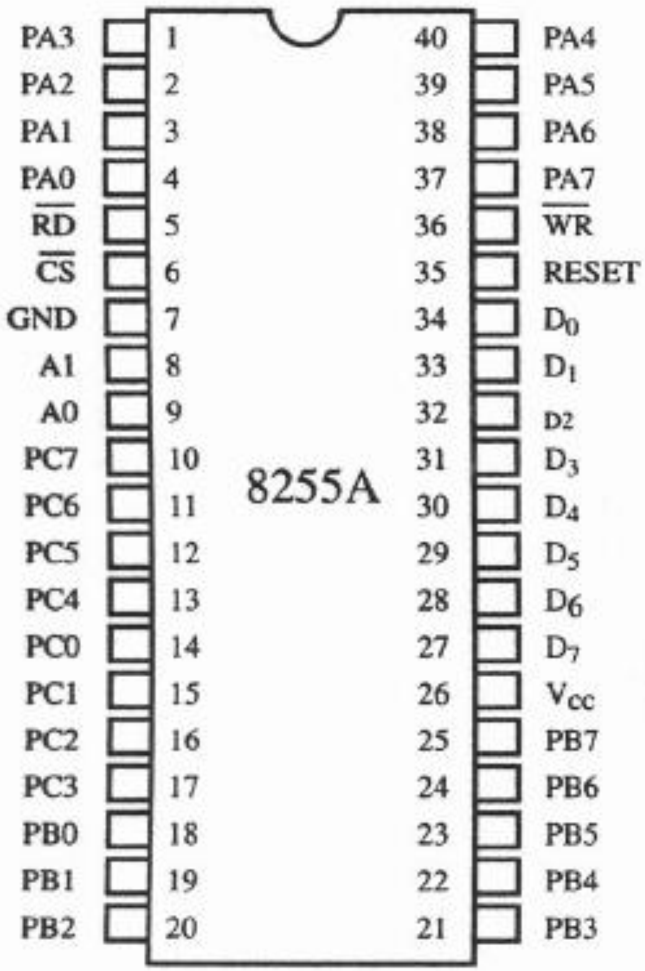
Basic Description of the 8255

- Each of the ports, **A** or **B**, can be programmed as an **8-bit input** or **output** port.
- Port **C** can be **divided in half**, with the topmost or bottommost four bits programmed as inputs or outputs.
- Individual bits of a particular port cannot be programmed.

Pin Configuration of the 8255

- The pin configuration of the 8255 is shown in Figure 1.
  - **GND**: System ground
  - **VCC**: System power
  - **RESET**: A high on this input clears the control register and all ports are set to the input mode.
  - **PA_{7-0}**: Port A bits
  - **PB_{7-0}**: Port B bits
  - **PC_{7-0}**: Port C bits
  - **D_{7-0}**: A bi-directional, tri-state data bus lines, connected to the system data bus.
  - **RD'**: A read input control, that is low during CPU read operations.
  - **WR'**: A write input control, that is low during CPU write operations.
  - **CS'**: A chip select control. A low on this input enables the 8255 to respond to RD' and WR' signals. RD' and WR' are ignored otherwise.
  - **A_{1-0}**: Address lines which in conjunction with RD' and WR', control the selection of one of the three ports or the control word registers as shown in Table 1.

Pin Configuration



Pin Names

D_7–D_0	Data Bus (Bidirectional)
RESET	Reset Input
$\overline{CS}$	Chip Select
$\overline{RD}$	Read Input
$\overline{WR}$	Write Input
A0, A1	Port Address
PA7–PA0	Port A (bit)
PB7–PB0	Port B (bit)
PC7–PC0	Port C (bit)
V_{cc}	+5V
GND	0V

Figure 1: Pin configuration of the 8255

Table 1: Selection of 8255 ports using address lines.

A_1	A_0	$\overline{RD}$	$\overline{WR}$	$\overline{CS}$	
					Input operation (READ)
0	0	0	1	0	Port A → data bus
0	1	0	1	0	Port B → data bus
1	0	0	1	0	Port C → data bus
					Output operation (WRITE)
0	0	1	0	0	Data bus → port A
0	1	1	0	0	Data bus → port B
1	0	1	0	0	Data bus → port C
1	1	1	0	0	Data bus → control
					Disable function
X	X	X	X	1	Data bus tristate
1	1	0	1	0	Illegal condition
X	X	1	1	0	Data bus tristate

Block Diagram of the 8255

- The block diagram of the 8255 is shown in Figure 2.
- Data Bus Buffer:
  - This **3-state bidirectional 8-bit buffer** is used to interface the 8255 to the system data bus.
  - Data is transmitted or received by the buffer upon execution of input or output instructions by the CPU.
  - Control words and status information are also transferred through the data bus buffer.
- Read/Write and Control Logic:
  - The function of this block is to manage all of the internal and external transfers of both Data and Control or Status words.
  - It accepts inputs from the CPU Address and Control busses and in turn, issues commands to both of the Control Groups.

Block Diagram of the 8255

- Group A and Group B Controls:
  - The functional configuration of each port is programmed by the **systems software**.
  - The CPU **outputs** a **control word** to the 8255.
  - The **control word** contains information such as **mode**, **bit set**, **bit reset**, etc., that initializes the functional configuration of the 82C55A.
  - Each of the Control blocks (**Group A** and **Group B**) accepts **commands** from the read/write control logic, receives **control words** from the internal data bus and issues the proper commands to its associated ports.
    - Control **Group A** - Port A and Port C upper (C_7-C_4)
    - Control **Group B** - Port B and Port C lower (C_3-C_0)
  - The control word register can be both **written** and **read** as shown in Table 1.

Block Diagram of the 8255

- Ports A, B, and C:
  - The 8255 contains **three 8-bit ports** (**A**, **B**, and **C**).
  - All can be configured in a wide variety of functional characteristics by the system software.

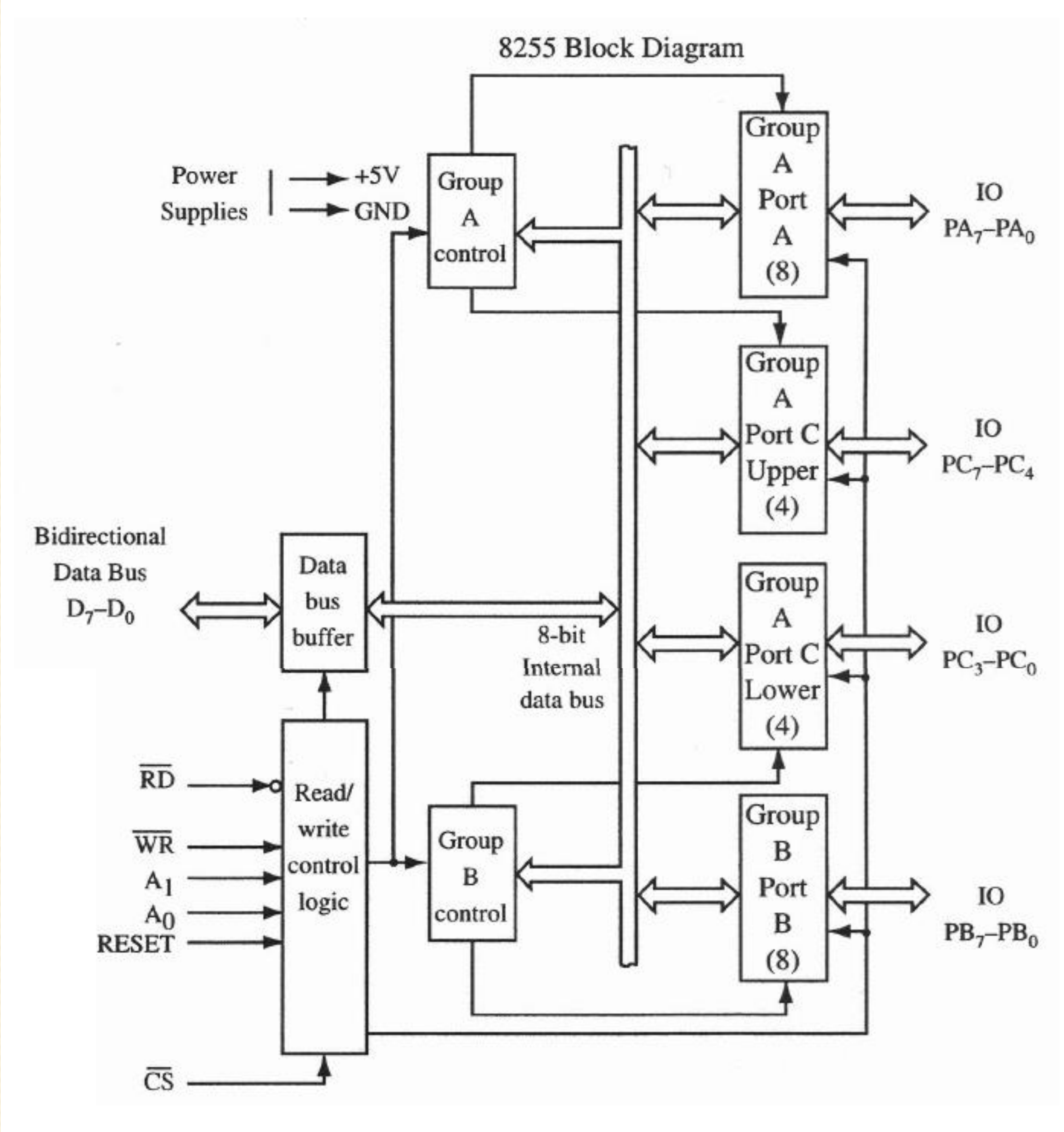


Figure 2: Block diagram of the 8255

Interfacing the 8255 to the 8086 Processor

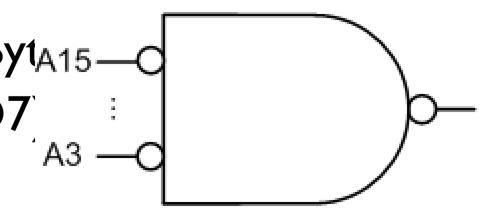
- **Example 1:** Show how to interface an 8255 chip to the low byte of the 8086 (D0-D7). Assume the following I/O address ports are used.

<u>Step (1):</u> Design the address decoding

A_{15}-A_{12}	A_{11}-A_{8}	A_{7}-A_{4}	A_{3}	A_{2}	A_{1}	A_{0}
0000	0000	0000	0	0	0	0
0000	0000	0000	0	0	1	0
0000	0000	0000	0	1	0	0
0000	0000	0000	0	1	1	0

Chip Select (CS') | Port Select (A_1 A_0) | Enable Even Byte (D0-D7)

Port Name	Port Address
Port A	00H
Port B	02H
Port C	04H
Control	06H



<u>Step(2):</u> Design control logic (IOW' & IOR')

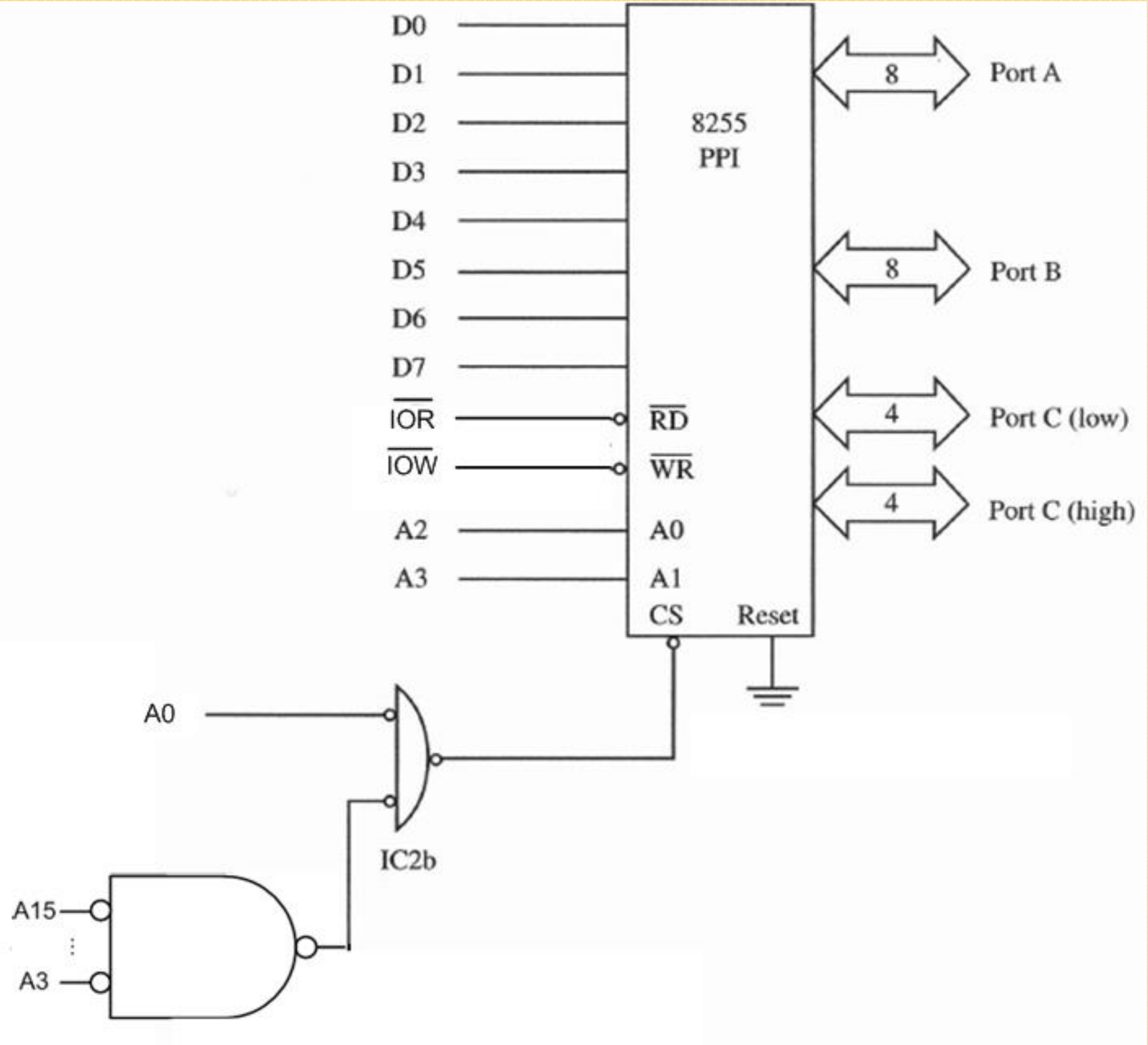


Figure 3: Interface of the 8255 in Example 1

Programming the 8255

- There are three basic modes of operation that can be selected by the system software:
  - **Mode 0:** Basic input/output
  - **Mode 1:** Strobed Input/output
  - **Mode 2:** Bi-directional Bus

Programming the 8255

- When the reset input of the 8255 goes "high" all ports will be set to the input mode with all 24 port lines held at a logic "one" level.
- After the reset is removed the 8255 can remain in the input mode with no additional initialization required.
- During the execution of the system program, any of the other modes may be selected by using a single output instruction.
- The modes for Port A and Port B can be separately defined, while Port C is divided into two portions.

Programming the 8255

- Figure 4 shows the format of the control byte used to program the 8255.
- There are two types of control bytes:
  - (a) When **bit 7 = 0**, a bit set/reset operation is indicated;
  - (b) When **bit 7 = 1**, any of the modes 0, 1, or 2 can be programmed.
- The ports in Group A can be programmed for any of modes 0, 1, or 2.
- The ports in Group B can only be programmed for modes 0 or 1.

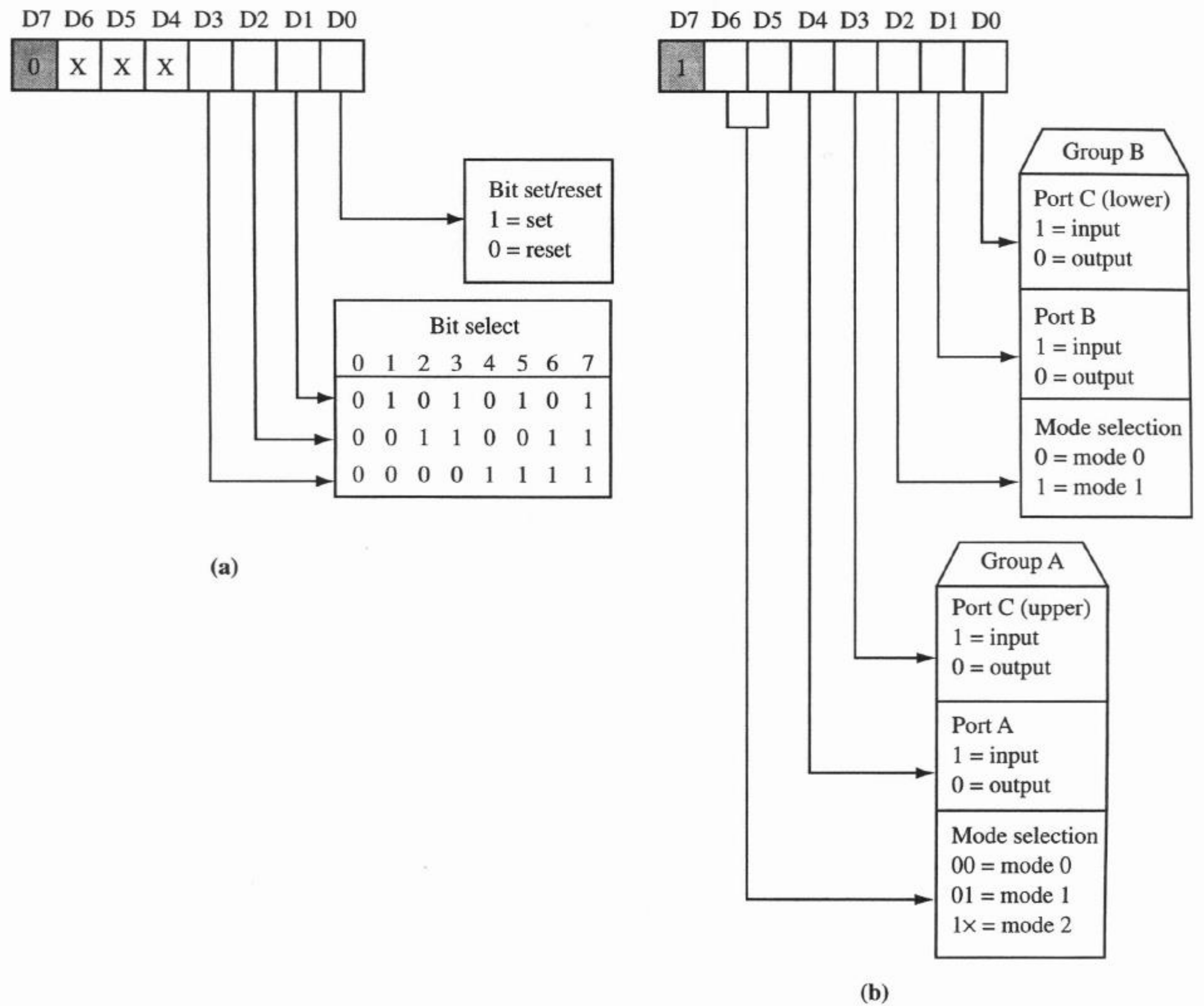


Figure 4: The format of the control byte of the 8255.

Programming the 8255

- **Example 2:** Write the 80x86 initialization routine required to program the 8255 in Figure 5 for mode 0, with port A as an output and ports B and C inputs
  - The control word is formed as:
    - 1 00 0 1 0 1 1 = 8BH
  - The program is as follows:
    - MOV AL,8BH ;Control byte to AL
    - OUT 6,AL ;Write to control port

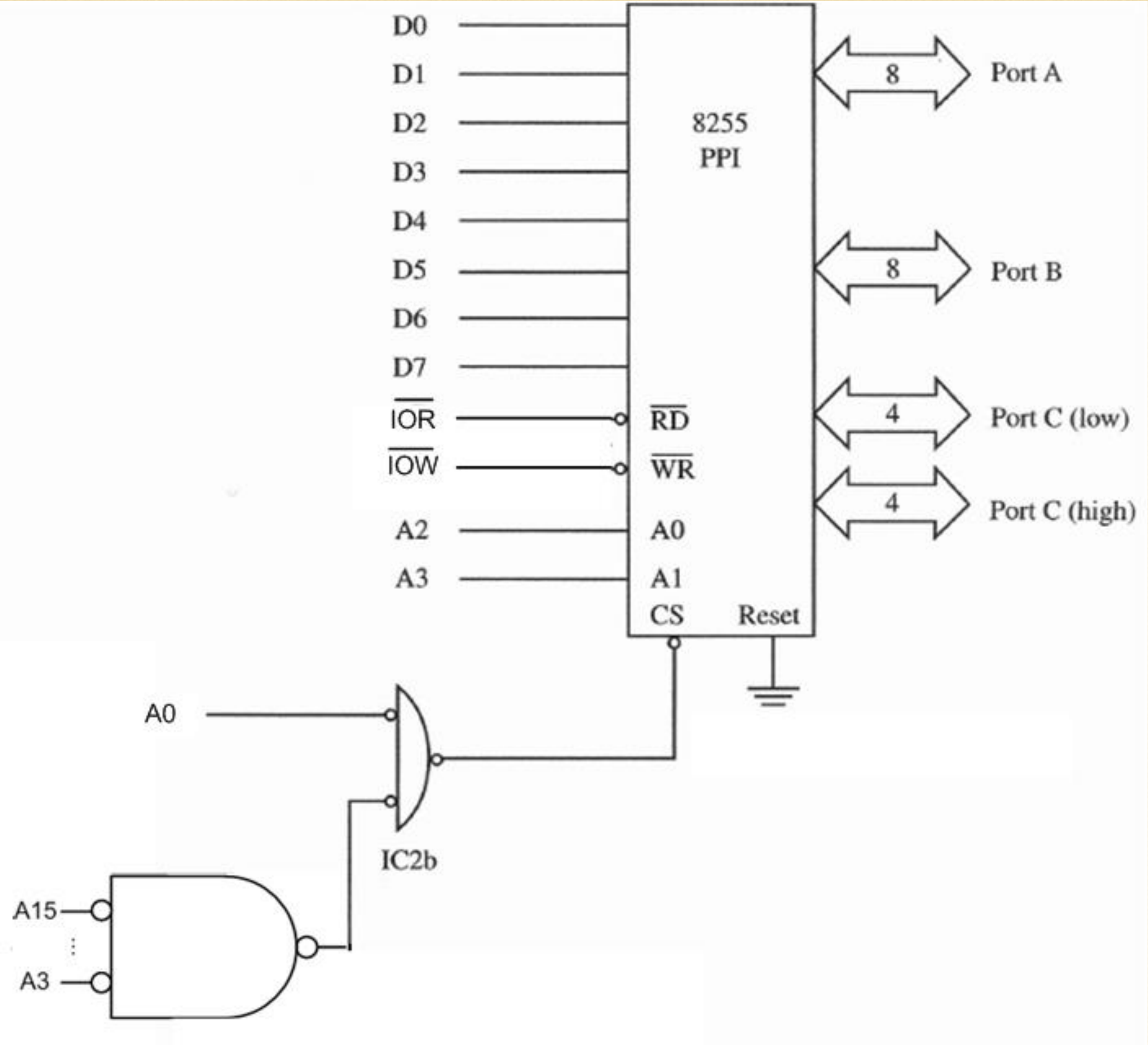


Figure 5: Circuit design of Example 2.

Programming the 8255

- **Example 3:** Write an 80x86 program to input a byte from port B of the PPI chip in pervious example and output this byte to port A of the same chip. Assume the chip has been programmed as in the previous example.

  - The program requires two instructions.
    - IN AL, 2 ; Get data from port B
    - OUT 0,AL ; Output the data to port A

Operating Modes of the 8255

- The 8255A can be programmed in three modes (0, 1, 2) as shown in Figure 6:
  - **Mode 0 (Basic I/O):** three simple I/O ports.
    - Ports A and B operate as either inputs or outputs.
    - Port C is divided into two 4-bit groups either of which can be operated as inputs or outputs.
  - **Mode 1 (Strobed I/O):** two hand shaking I/O ports.
    - Ports A and B operate as either inputs or outputs as in mode 0
    - Port C is used for handshaking and control.
  - **Mode 2 (Strobed Bidirectional I/O):** a bidirectional I/O port with five hand shaking signals.
    - Port A is bidirectional (both input and output).
    - Port C is used for handshaking.
    - Port B is not used.
- These modes can also be intermixed. For example, port A can be programmed to operate in mode 2, while port B operates in mode 0.

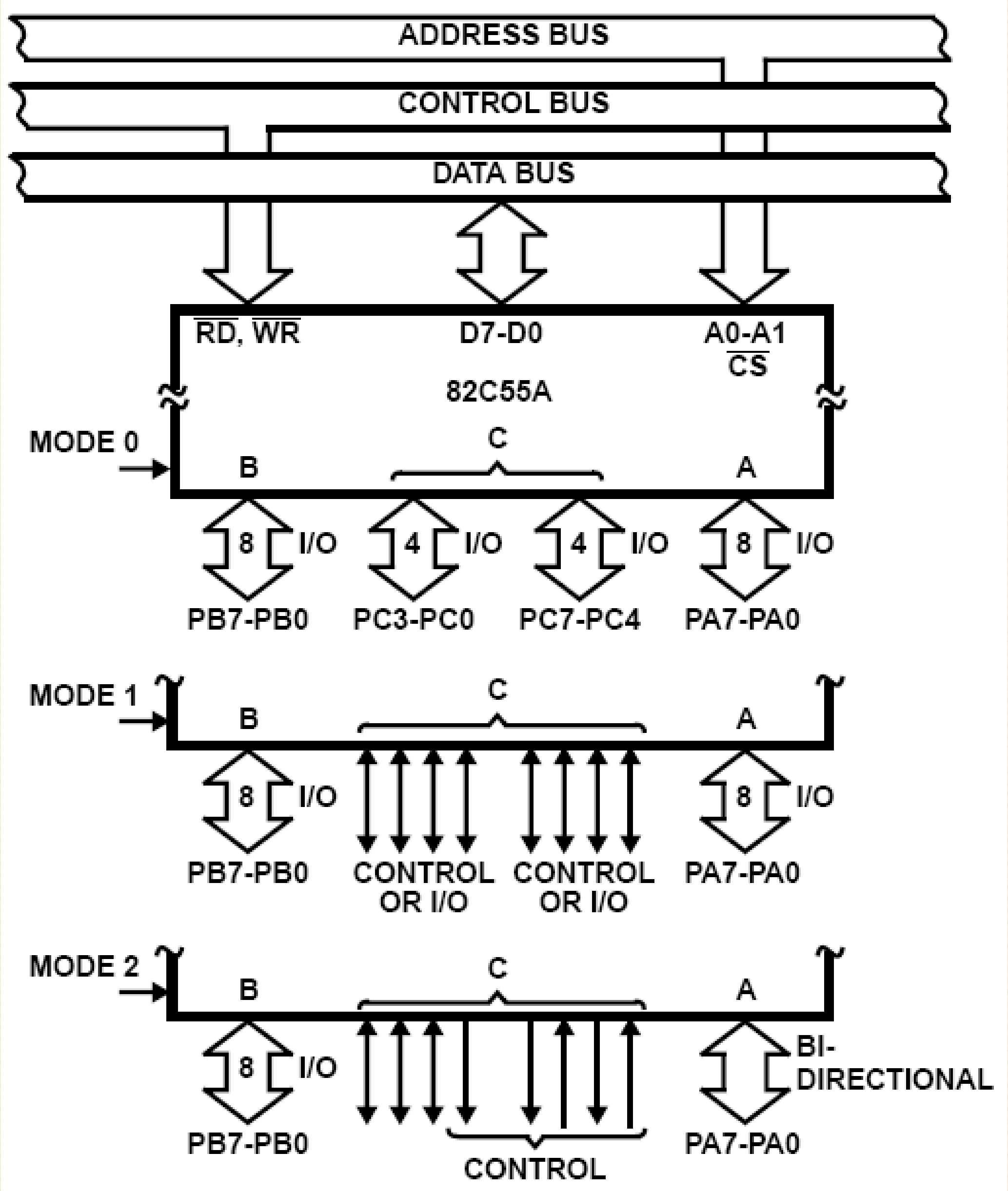


Figure 6: The three basic modes of the 8255.

Operating Modes of the 8255

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Mode 0 (Basic Input / Output)

- This mode provides simple input and output operations for each of the three ports.
- No handshaking is required, data is simply written to or read from a specific port.
- The basic features of this mode are:
  - Two 8-bit ports and two 4-bit ports
  - Any Port can be input or output
  - Outputs are latched
  - Input are not latched
  - 16 different input / output configurations possible as shown in Table 2.

Table 2: Mode 0 port definition.

A		B		GROUP A		#	GROUP B	
D4	D3	D1	D0	PORT A	PORT C (Upper)		PORT B	PORT C (Lower)
0	0	0	0	Output	Output	0	Output	Output
0	0	0	1	Output	Output	1	Output	Input
0	0	1	0	Output	Output	2	Input	Output
0	0	1	1	Output	Output	3	Input	Input
0	1	0	0	Output	Input	4	Output	Output
0	1	0	1	Output	Input	5	Output	Input
0	1	1	0	Output	Input	6	Input	Output
0	1	1	1	Output	Input	7	Input	Input
1	0	0	0	Input	Output	8	Output	Output
1	0	0	1	Input	Output	9	Output	Input
1	0	1	0	Input	Output	10	Input	Output
1	0	1	1	Input	Output	11	Input	Input
1	1	0	0	Input	Input	12	Output	Output
1	1	0	1	Input	Input	13	Output	Input
1	1	1	0	Input	Input	14	Input	Output
1	1	1	1	Input	Input	15	Input	Input

Operating Modes of the 8255

Mode 1 (Strobed Input / Output)

- This mode provides a means for transferring I/O data to or from a specified port in conjunction with strobes or "hand shaking" signals.
- In this mode, port A and port B use the lines on port C to generate or accept these "hand shaking" signals.
- The basic features of this mode are:
    - Two Groups (Group A and Group B).
    - Each group contains one 8-bit port and one 4-bit control/data port.
    - The 8-bit data port can be either input or output.
    - Both inputs and outputs are latched.
    - The 4-bit port is used for control and status of the 8-bit port.

Operating Modes of the 8255

Mode 1 (Strobed Input / Output)

- Figure 7 shows the control signals for input configuration.
- **STB (Strobe Input)**
  - A "low" on this input loads data into the input latch.
- **IBF (Input Buffer Full F/F)**
  - A "high" on this output indicates that the data has been loaded into the input latch.
  - IBF is set by STB' input being low and is reset by the rising edge of the RD' input.
- **INTR (Interrupt Request)**
  - A "high" on this output can be used to interrupt the CPU when and input device is requesting service.
  - INTR is set by the condition: STB is a "one", IBF is a "one" and INTE is a "one".
  - It is reset by the falling edge of RD.
  - This procedure allows an input device to request service from the CPU by simply strobing its data into the port.
- **INTE A:** Controlled by bit set/reset of PC4.
- **INTE B:** Controlled by bit set/reset of PC2.

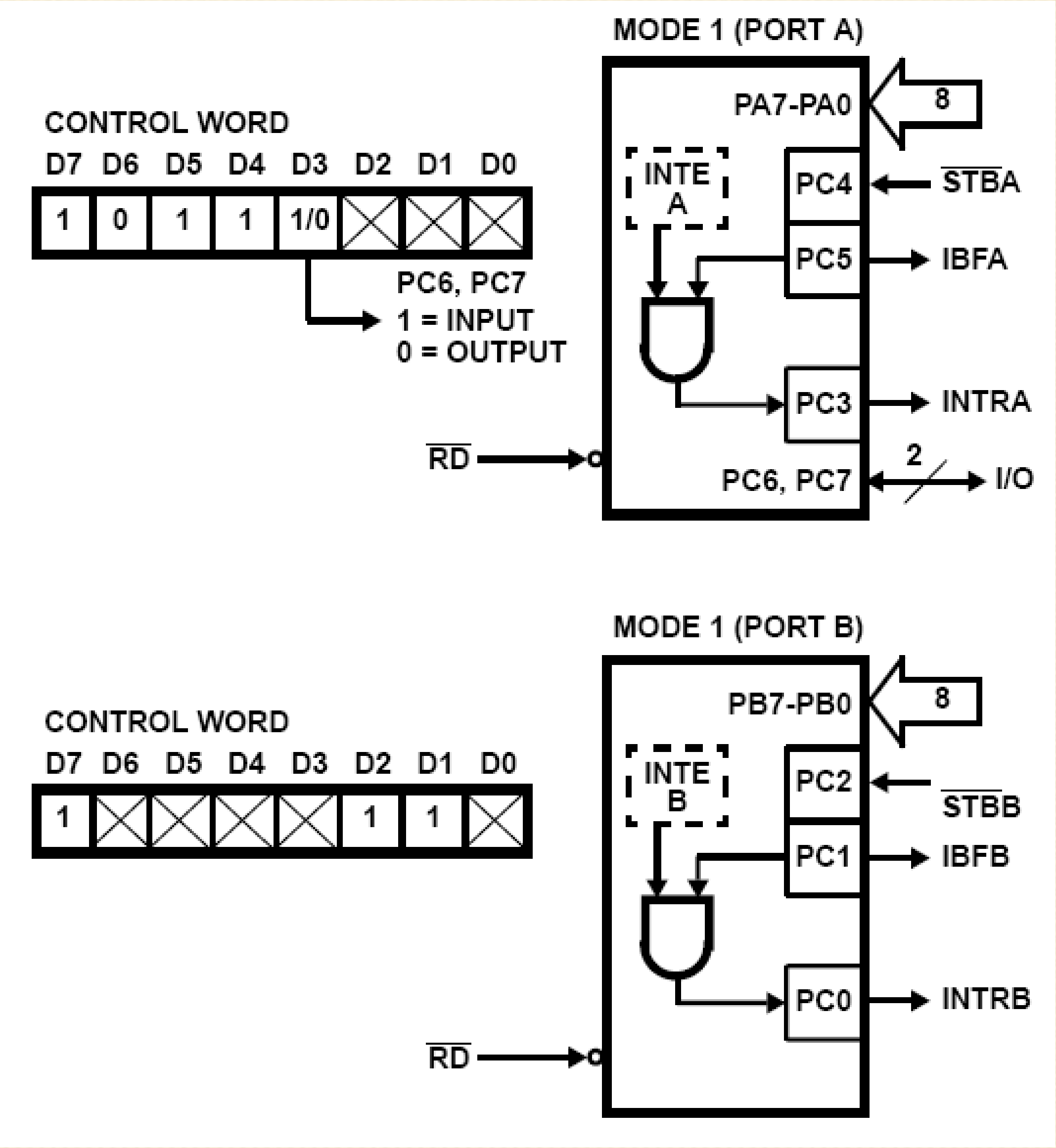


Figure 7: Mode 1 input.

Operating Modes of the 8255

Mode 1 (Strobed Input / Output)

- Figure 8 shows the control signals for output configuration.
- **OBF - Output Buffer Full F/F:**
  - The OBF' output will go "low" to indicate that the CPU has written data out to be specified port.
  - The OBF' F/F will be set by the rising edge of the WR input and reset by ACK input being low.
- **ACK - Acknowledge Input):**
  - A "low" on this input informs the 82C55A that the data from Port A or Port B is ready to be accepted.
  - A response from the peripheral device indicating that it is ready to accept data
- **INTR - (Interrupt Request):**
  - A "high" on this output can be used to interrupt the CPU when an output device has accepted data transmitted by the CPU.
  - INTR is set when ACK is a "one", OBF is a "one" and INTE is a "one".
  - It is reset by the falling edge of WR.
- **INTE A:** Controlled by bit set/reset of PC6.
- **INTE B:** Controlled by bit set/reset of PC2.

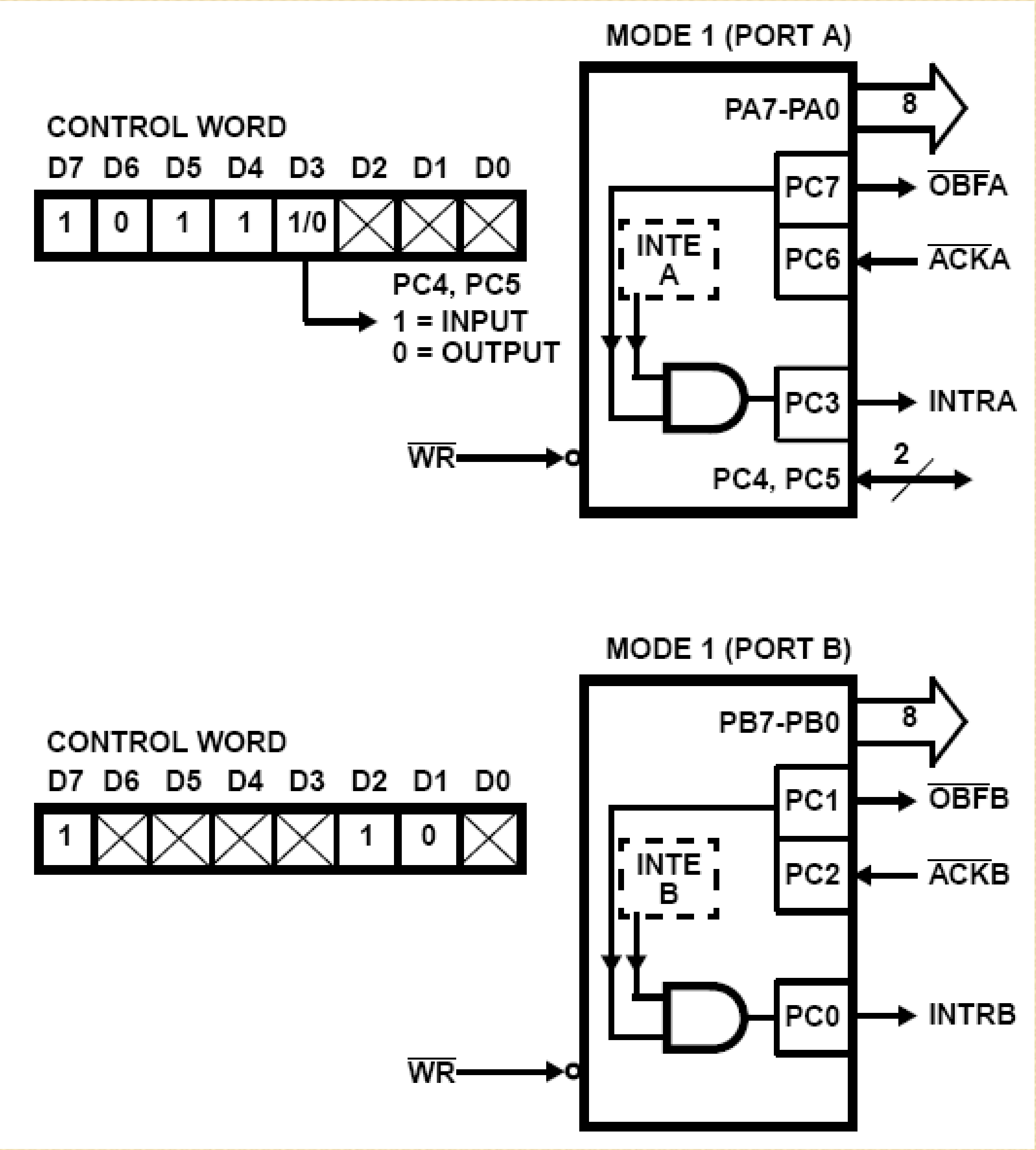


Figure 8: Mode 1 output.

Operating Modes of the 8255

Mode 2 (Strobed Bidirectional Input / Output)

- This mode provides a means for communicating with a peripheral device or structure on a single 8-bit bus for both transmitting and receiving data (bidirectional bus I/O).
- In this mode, port A uses the lines on port C to generate or accept these "hand shaking" signals.
- The basic features of this mode are:
  - Used in Group A only.
  - One 8-bit, bi-directional bus Port (Port A) and a 5-bit control Port (Port C)
  - Both inputs and outputs are latched.
  - The 5-bit control port (Port C) is used for control and status for the 8-bit, bi-directional bus port (Port A)

Operating Modes of the 8255

Mode 2 (Strobed Bidirectional Input / Output)

- Figure 9 shows the control signals for mode 2 configurations.
- Input Operations:
  - **STB' -** (Strobe Input)**:** A "low" on this input loads data into the input latch.
  - **IBF** - (Input Buffer Full F/F)**:** A "high" on this output indicates that data has been loaded into the input latch.
  - **INTE 2** - (The INTE flip-flop associated with IBF)**:** Controlled by bit set/reset of PC4.
  - **INTR -** (Interrupt Request)**:** A high on this output can be used to interrupt the CPU for both input or output operations.

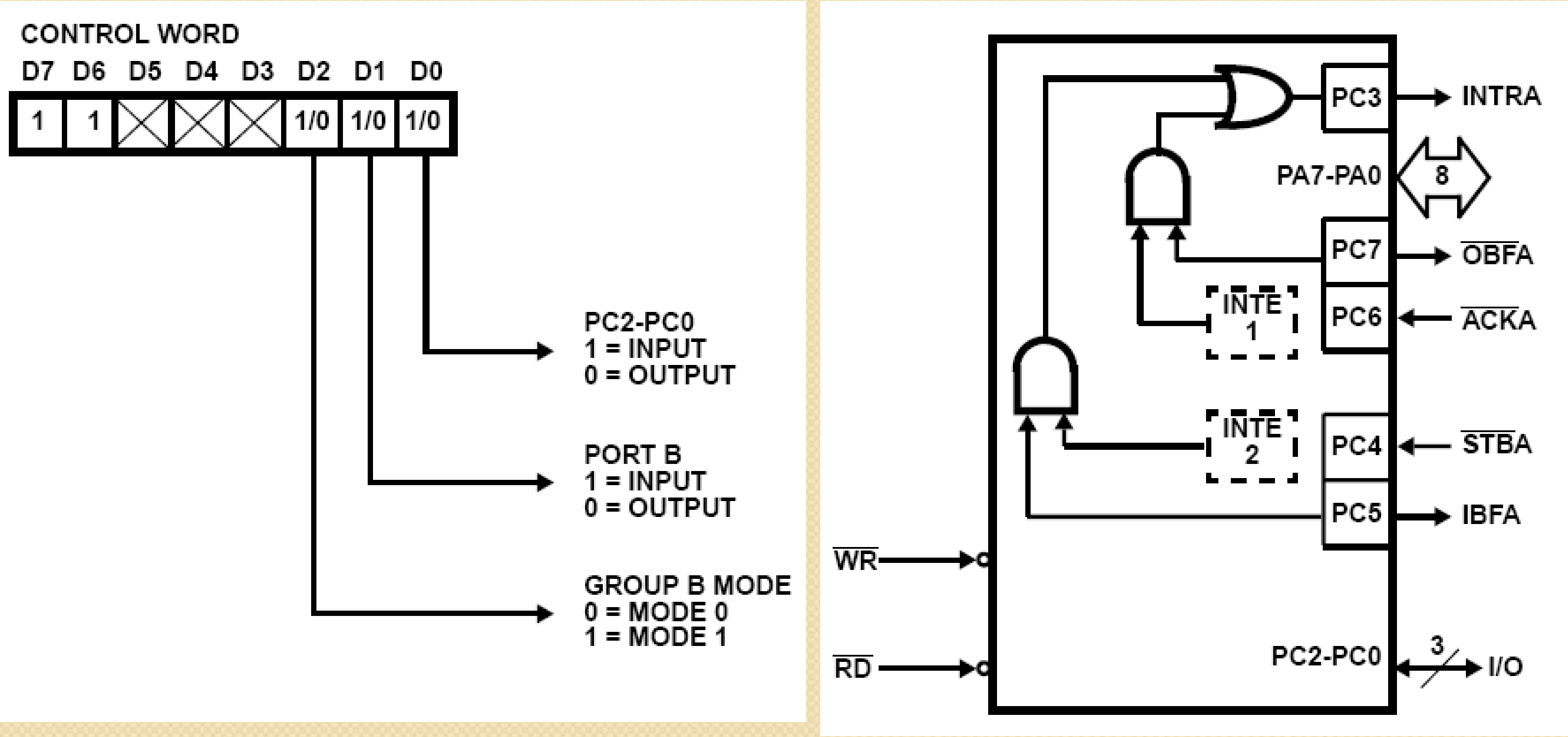


Figure 9: Mode 2 control signals.

Operating Modes of the 8255

Mode 2 (Strobed Bidirectional Input / Output)

- <u>Input Operations</u>:
  - **STB' -** (Strobe Input)**:** A "low" on this input loads data into the input latch.
  - **IBF** - (Input Buffer Full F/F)**:** A "high" on this output indicates that data has been loaded into the input latch.
  - **INTE 2** - (The INTE flip-flop associated with IBF)**:** Controlled by bit set/reset of PC4.
  - **INTR -** (Interrupt Request)**:** A high on this output can be used to interrupt the CPU for both input or output operations.
- <u>Output Operations</u>:
  - **OBF'** - (Output Buffer Full): The OBF output will go "low" to indicate that the CPU has written data out to port A.
  - **ACK'** - (Acknowledge): A "low" on this input enables the three-state output buffer of port A to send out the data. Otherwise, the output buffer will be in the high impedance state.
  - **INTE 1** - (The INTE flip-flop associated with OBF): Controlled by bit set/reset of PC4.

Basic DMA concept

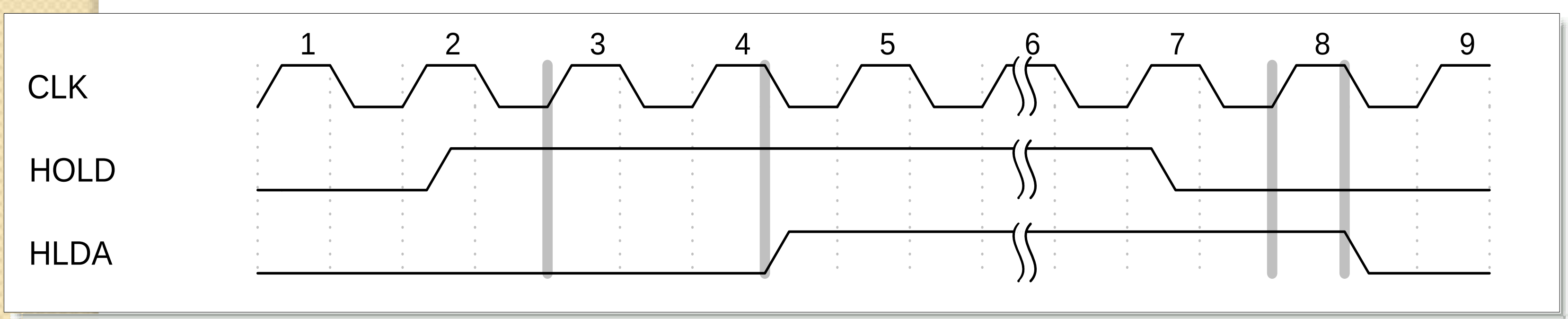
- **Direct memory access** (**DMA**) is a feature of modern computer systems that allows certain hardware subsystems to read/write data to/from memory without microprocessor intervention, allowing the processor to do other work.
- Used in disk controllers, video/sound cards etc, or between memory locations.
- Typically, the CPU initiates DMA transfer, does other operations while the transfer is in progress, and receives an interrupt from the DMA controller once the operation is

BASIC DMA TERMINOLOGY

- *DMA channel*: system pathway used by a device to transfer information directly to and from memory. There are usually 8 in a computer system
- *DMA controller*: dedicated hardware used for controlling the DMA operation
- *Single-cycle mode:* DMA data transfer is done one byte at a time
- *Burst-mode:* DMA transfer is finished when all data has been moved

DMA pins and timing

- x86 Interrupt Pins
  - HOLD: DMA request.
    - Sampled in the middle of any clocking cycle
  - HLDA: DMA acknowledge signal.
    - The address, data and control buses are set to high-Z, so the I/O devices can control the system bus



DMA on the 8086 Microprocessor

- The I/O device asserts the appropriate DRQ signal for the channel.
- The DMA controller will enable appropriate channel, and ask the CPU to release the bus so that the DMA may use the bus. The DMA requests the bus by asserting the HOLD signal which goes to the CPU.
- The CPU detects the HOLD signal, and will complete executing the current instruction. Now all of the signals normally generated by the CPU are placed in a tri-stated condition (neither high or low) and then the CPU asserts the HLDA signal which tells the DMA controller that it is now in charge of the bus.
- The CPU may have to wait (hold cycles).
- DMA activates its -MEMR, -MEMW, -IOR, -IOW output signals, and the address outputs from the DMA are set to the target address, which will be used to direct the byte that is about to transferred to a specific memory location.

DMA on the 8086 Microprocessor

- The DMA will then let the device that requested the DMA transfer know that the transfer is commencing by asserting the -DACK signal.
- The peripheral places the byte to be transferred on the bus Data lines.
- Once the data has been transferred, The DMA will de-assert the -DACK2 signal, so that the FDC knows it must stop placing data on the bus.
- The DMA will now check to see if any of the other DMA channels have any work to do. If none of the channels have their DRQ lines asserted, the DMA controller has completed its work and will now tri-state the -MEMR, -MEMW, -IOR, -IOW and address signals.
- Finally, the DMA will de-assert the HOLD signal. The CPU sees this, and de-asserts the HOLDA signal. Now the CPU resumes control of the buses and address lines, and it resumes executing instructions and accessing main memory and the peripherals.

EXAMPLE

- Assuming that a DMA initialization has an overhead of 10 cycles, while a CPU transfer to/from memory requires 4 cycles (no wait states required), compare a DMA and a CPU transfer from one memory location to another of
  - One byte of data
  - A block of 1Kbytes in burst mode
  - A block of 64Kbytes in burst mode

The 8237 DMA controller

- Supplies memory and I/O with control signals and addresses during DMA transfer
- 4-channels (expandable)
  - 0: DRAM refresh
  - 1: Free
  - 2: Floppy disk controller
  - 3: Free
- 1.6MByte/sec transfer rate
- 64 KByte section of memory address capability with single programming
- "fly-by" controller (data does not pass through the DMA-only memory to I/O transfer capability)

8237 pins

- CLK: System clock
- CS´: Chip select (decoder output)
- RESET: Clears registers, sets mask register
- READY: 0 for inserting wait states
- HLDA: Signals that the μp has relinquished buses
- DREQ3 – DREQ0: DMA request input for each channel
- DB7-DB0: Data bus pins
- IOR´: Bidirectional pin used during programming

and during a DMA write cycle
- IOW´: Bidirectional pin used during programming

and during a DMA read cycle
- EOP´: End of process is a bidirectional signal used as input to terminate a DMA process or as output to signal the end of the DMA transfer
- A3-A0: Address pins for selecting internal registers
- A7-A4: Outputs that provide part of the DMA transfer address
- HRQ: DMA request output
- DACK3-DACK0: DMA acknowledge for each channel.
- AEN: Address enable signal
- ADSTB: Address strobe
- MEMR´: Memory read output used in DMA read cycle
- MEMW´: Memory write output used in DMA write cycle

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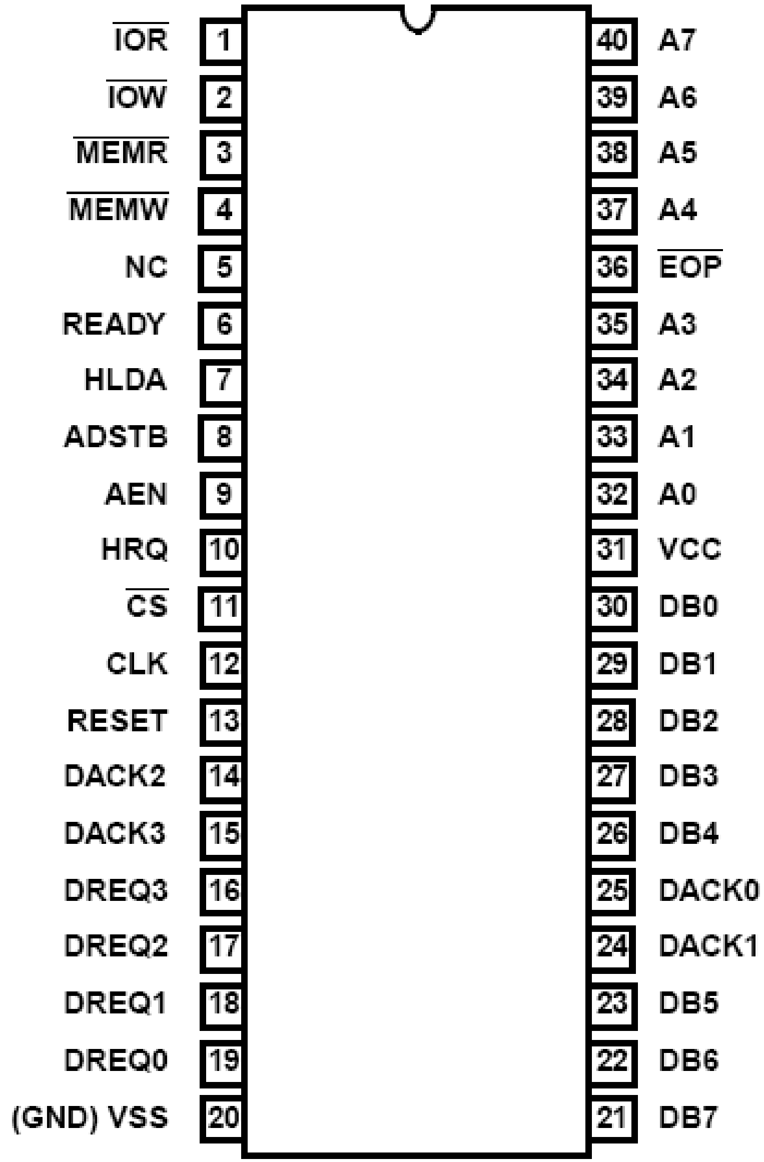
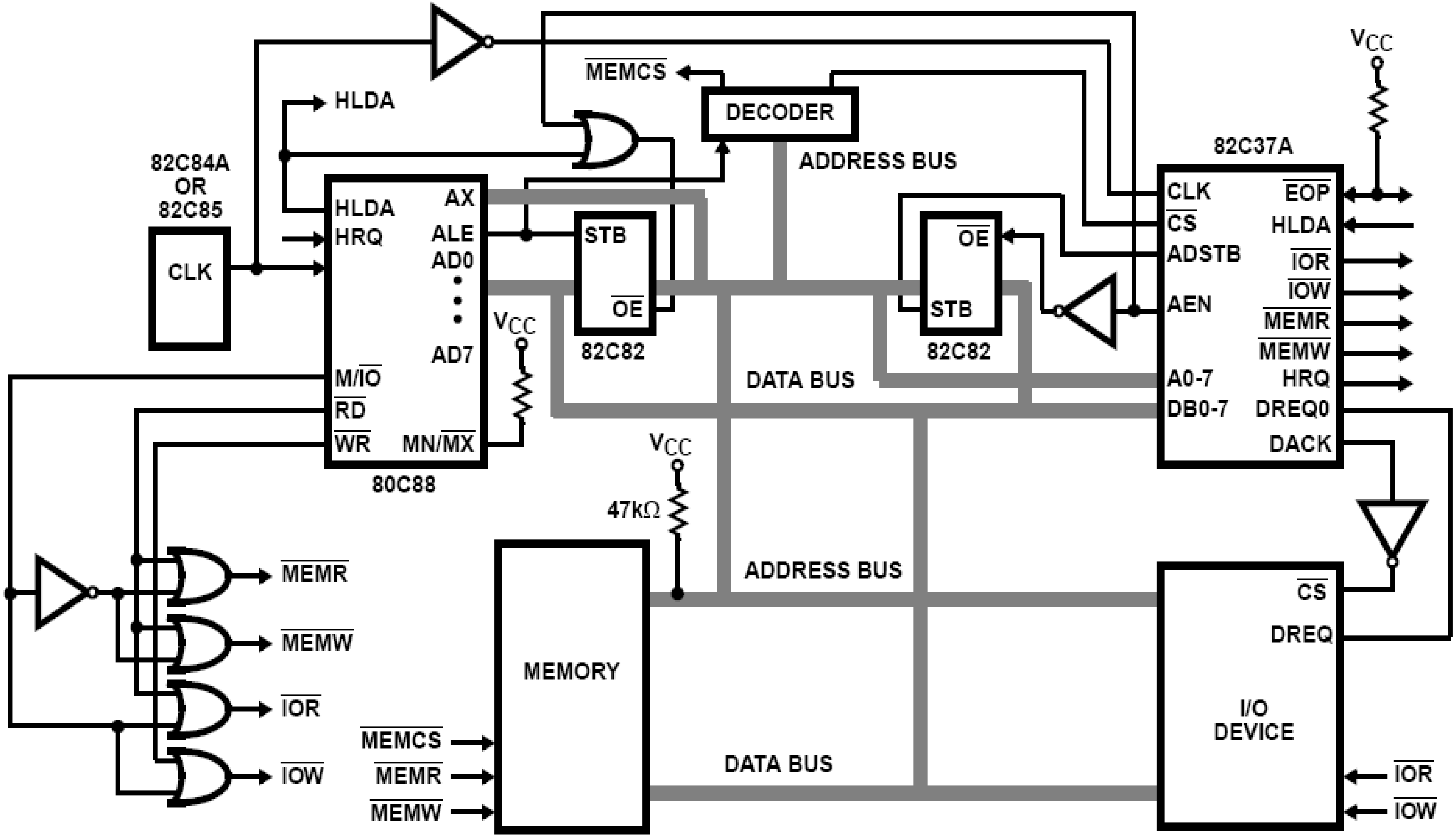
Diagrar





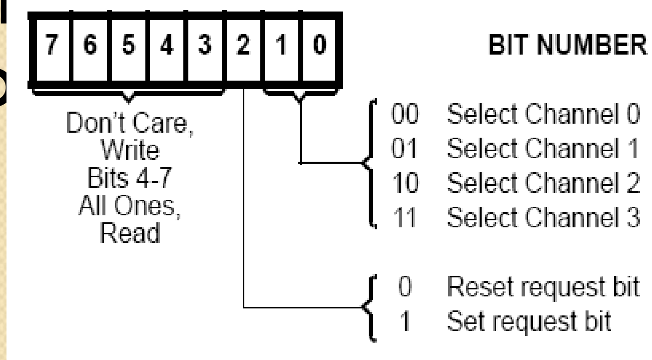
DMA ARCHITECTURE

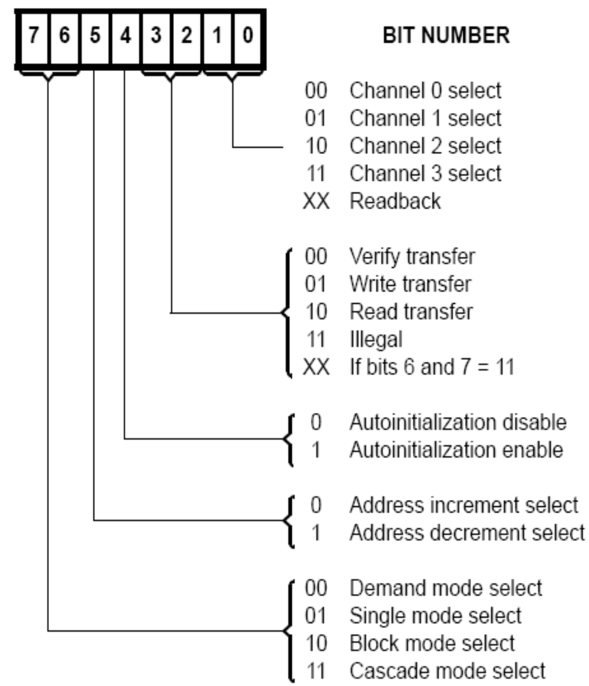
8237 registers

- CAR (Current Address Register): holds the 16-bit memory address used for the DMA transfer (one for each channel), either incremented or decremented during the operation
- CWCR (Current Word Count Register): Programs a channel for the number of bytes (up to 64K) transferred during a DMA operation
- BA (Base Address) and WC (Word

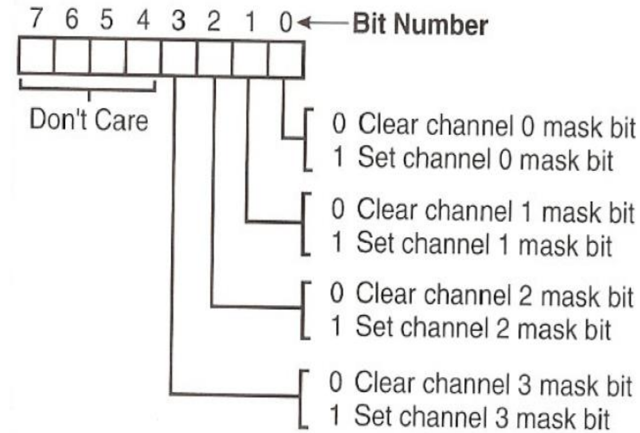
- MR (Mode Register):

- Programs the mode of operation for a channel (one fo

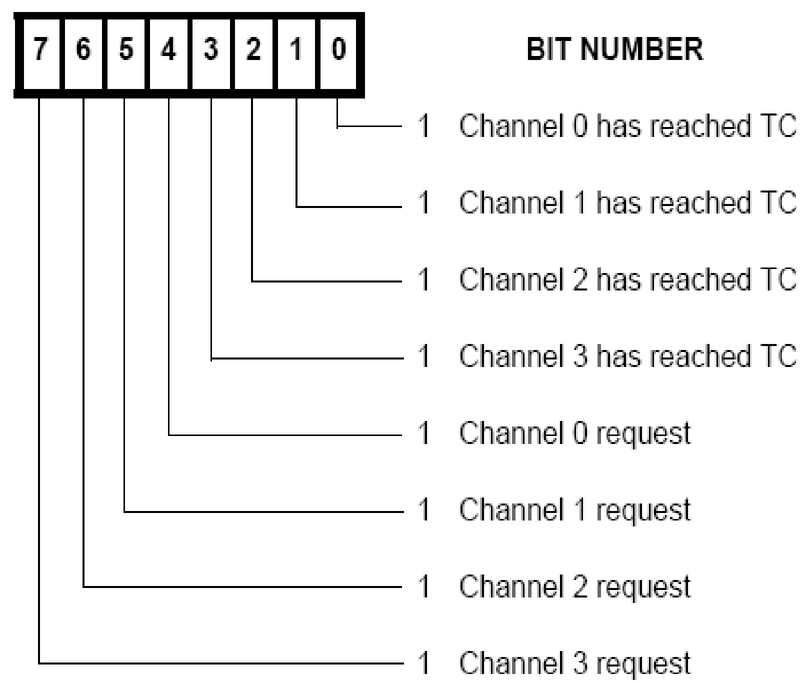




- MR (Mask Register):



- SR (Status Register): Shows the status of each DMA channel



8237 Software commands

OPERATION	A3	A2	A1	A0	$\overline{IOR}$	$\overline{IOW}$
Read Status Register	1	0	0	0	0	1
Write Command Register	1	0	0	0	1	0
Read Request Register	1	0	0	1	0	1
Write Request Register	1	0	0	1	1	0
Read Command Register	1	0	1	0	0	1
Write Single Mask Bit	1	0	1	0	1	0
Read Mode Register	1	0	1	1	0	1
Write Mode Register	1	0	1	1	1	0
Set First/Last F/F	1	1	0	0	0	1
Clear First/Last F/F	1	1	0	0	1	0
Read Temporary Register	1	1	0	1	0	1
Master Clear	1	1	0	1	1	0
Clear Mode Reg. Counter	1	1	1	0	0	1
Clear Mask Register	1	1	1	0	1	0
Read All Mask Bits	1	1	1	1	0	1
Write All Mask Bits	1	1	1	1	1	0

8237 Software commands

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- **Clear First/Last Flip-Flop -** This command is executed prior to writing or reading new address or word count information to the 82C37. This command initializes the flipflop to a known state (low byte first) so that subsequent accesses to register contents by the microprocessor will address upper and lower bytes in the correct sequence.
- **Set First/Last Flip-Flop -** This command will set the flip-flop to select the high byte first on read and write operations to address and word count registers.
- **Master Clear -** This software instruction has the same effect as the hardware Reset. The Command, Status, Request, and Temporary registers, and Internal First/Last Flip-Flop and mode register counter are cleared and the Mask register is set. The 82C37A will enter the idle cycle.
- **Clear Mask Register -** This command clears the mask bits of all four channels, enabling them to accept DMA requests.
- **Clear Mode Register Counter -** Since only one address location is available for reading the Mode registers, an

8237 block diagram